

Predicting Cross-Asset Stablecoin Contagion with Temporal Graph Neural Networks

Anonymous Author(s)
Columbia University, MAFN
New York, USA

Abstract

On 10 March 2023, the stablecoin USDC—a token designed to always trade at exactly one US dollar—fell to \$0.88 after its issuer disclosed that \$3.3B of its cash reserves were trapped at the failed Silicon Valley Bank. Within hours, *other* dollar stablecoins wobbled too. This paper asks a concrete, operational question: when one stablecoin breaks its dollar peg, *which other stablecoins will follow, and how soon?* We treat this as a prediction problem on a network that evolves through time, using real minute-by-minute price data from two large exchanges across seven crisis episodes in 2022–2023. We compare a full ladder of models—from simple baselines to gradient-boosted trees to temporal graph neural networks (GNNs)—under an evaluation protocol carefully designed to prevent the model from “cheating” by seeing overlapping crises. Two findings stand out. First, contagion is essentially *unpredictable* at horizons of a few hours, but becomes predictable at a one-day horizon, where a graph-attention network (GAT) beats every non-graph model by 0.18 in precision–recall AUC. Second, an ablation that deletes the network’s edges shows that this advantage comes specifically from the *relationships between stablecoins*, not merely from a more flexible model. We deliver an interpretable ranking of the most systemically central venues in each crisis, designed as the input to downstream risk and policy analysis.

Keywords

stablecoins, financial contagion, temporal graph neural networks, graph attention, systemic risk, leakage-safe evaluation

1 Introduction

1.1 What is a stablecoin, and what is a depeg?

A *stablecoin* is a cryptocurrency token engineered to hold a constant value, almost always one US dollar. Users treat \$1 of a stablecoin as interchangeable with \$1 of cash, which is why stablecoins now settle trillions of dollars of transactions and underpin much of decentralized finance. The largest fiat-backed stablecoins (USDC, USDT, formerly BUSD) promise that every token is backed one-for-one by cash and short-term government debt; crypto-collateralized coins (DAI, FRAX) are instead backed by a basket of other crypto assets, often *including other stablecoins*.

When a stablecoin’s market price moves away from \$1, it is said to *depeg*. A depeg is the crypto analogue of a bank run: holders fear the token is no longer fully backed and rush to sell or redeem it, pushing the price further down. Because regulators now treat stablecoins as systemically important—both the US GENIUS Act and the EU’s MiCA regulation make reserve adequacy and contagion containment central concerns [2, 4]—understanding how one depeg spreads to others is a question with direct policy stakes.

1.2 A concrete example: the SVB weekend

On the weekend of 10–12 March 2023, Silicon Valley Bank failed. Circle, the issuer of USDC, disclosed that \$3.3B of the cash backing USDC was stuck at SVB. The market could not tell whether that money would be recovered, and USDC—supposedly worth exactly \$1—traded as low as \$0.88. Crucially, the panic did not stay contained. **DAI** fell too, because a large share of DAI’s collateral was *itself* USDC; **FRAX**, roughly 90% USDC-backed, also dropped; and even unrelated coins such as **TUSD** and **Pax Dollar (USDP)** wobbled as traders fled stablecoins indiscriminately. Figure 1 shows the real minute-level paths. This is *contagion*: a shock to one node of a financial network propagating to others.

1.3 The prediction problem and our contributions

We ask: given the state of the stablecoin market at time t , can we predict which currently healthy stablecoins will depeg within a horizon Δ ? This is a forecasting problem on a *temporal graph*—a network whose nodes are trading venues, whose edges capture which venues move together and in what order, and whose structure changes minute by minute. Graph neural networks are the natural model class for learning on such structured, evolving data, and have recently been applied to related on-chain prediction tasks such as liquidity-pool bridge swaps on Uniswap [7]. Our contributions are:

- (1) A contagion predictor trained on **real** multi-venue, minute-level data, evaluated under a **leakage-safe** protocol that we show is essential (§3).
- (2) A **lead-time** finding: contagion is unpredictable at hours but predictable at a day, established with metrics that are not inflated by class imbalance (§5).
- (3) An **ablation** that isolates how much of the model’s skill comes from the network structure itself, and shows the answer is attention-specific (§5).
- (4) An interpretable, exportable **hub ranking** that names the most central venues in each crisis (§6).

2 Background and related work

Graph neural networks in one paragraph. A graph neural network learns a representation for each node by repeatedly mixing the node’s own features with those of its neighbours—“message passing.” **GraphSAGE** [3] averages neighbour information; **graph attention networks (GAT)** [5] instead *learn how much weight* to give each neighbour, which matters when some relationships are far more informative than others. A *temporal graph* applies this at each time step on a network that itself changes over time.

Prior work. GNNs have been used to predict on-chain events such as cross-pool bridge swaps in Uniswap v3 [7], but evaluate

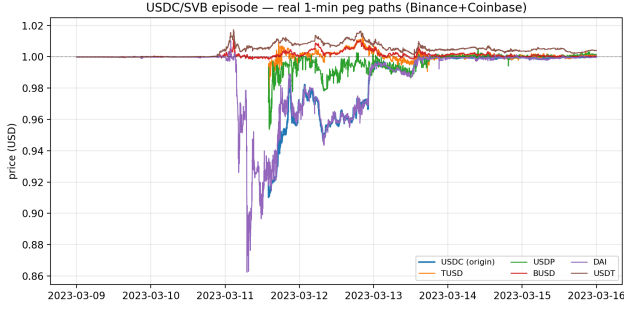


Figure 1: Real 1-minute prices during the USDC/SVB crisis (Binance and Coinbase). USDC and several peers fall to \$0.86–0.95 on 11 March 2023 and recover by the 13th. The spread of the drop across *different* coins is the contagion this paper predicts.

prediction, not which nodes *cause* propagation. Tree ensembles such as XGBoost [1] are the strong tabular baseline we must beat to justify a graph model. We follow this literature in framing contagion as a temporal node-classification task, and depart from it by (i) working cross-asset on the deepest centralized venues, where minute-level data is cleanest, and (ii) treating leakage control as a first-class design constraint.

3 Data and task formulation

Data. We collect real 1-minute open/high/low/close/volume bars from **Binance** (for USDC, TUSD, USDP, FRAX, BUSD, FDUSD, and the failed algorithmic coin UST, all quoted against USDT) and **Coinbase** (for DAI and USDT, quoted against the US dollar). Because Binance prices are quoted in USDT rather than dollars, we multiply them by the Coinbase USDT/USD price so that a wobble in USDT itself is not hidden by the choice of quote currency. We assemble seven historical crisis episodes: the Terra/UST collapse and the secondary USDT scare (May 2022), the FTX-driven DAI stress (Nov 2022), the regulatory BUSD wind-down (Feb 2023), the USDC and FRAX SVB depegs (Mar 2023), and the 2018 USDT scare.

Nodes, samples, and labels. A *node* is an (asset, venue) pair, e.g. “USDC on Binance.” We sample the market once per hour; each (hourly snapshot, node) pair is one training example. The *label* is contagion *onset*: node j enters a *new* sustained stress period—its price more than an asset-class-specific threshold (25 basis points for fiat-backed, 75 for crypto-backed) away from \$1 for at least ten consecutive minutes—within the next Δ minutes, given that it was calm at time t . The coin that triggered the episode (the “origin”) is excluded, because we want to predict *spread*, not the initial shock. We use four horizons: 30 minutes, 1 hour, 4 hours, and 24 hours.

Why leakage control is the crux. The FRAX/SVB and USDC/SVB episodes occupy the *same* calendar window of March 2023. If one is placed in the training set and the other in the test set, a model can memorize the idiosyncratic market conditions of that weekend and appear near-perfect—we observed XGBoost reaching a PR-AUC of 1.0, an obvious artifact. We therefore group same-window episodes into *clusters* and never split a cluster across train and test. We report

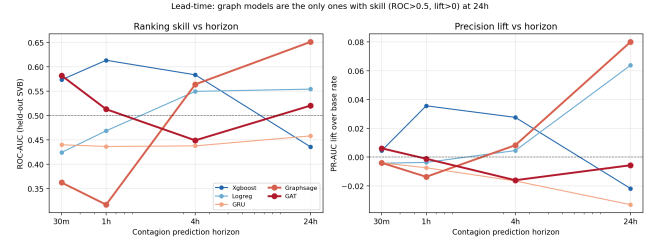


Figure 2: Lead-time. Left: ranking skill (ROC-AUC) versus horizon—only the graph models beat chance (0.5, dashed) at 24h. Right: precision lift over the base rate. Per-node tabular models stay near zero skill.

two protocols: a held-out SVB cluster, and leave-one-cluster-out (LOCO), in which each cluster takes a turn as the test set. This single discipline is what separates an honest result from a misleading one.

4 Models

We evaluate a *ladder* of models on identical samples, from simplest to most expressive: (1) *majority* and *persistence* baselines; (2) *logistic regression* and XGBoost [1] on each node’s own features; (3) a GRU recurrent network on per-node feature sequences; and (4) two GNNs, GraphSAGE and GAT, on the temporal graph. Node features are standard market-microstructure signals—price deviation from \$1, realized volatility over 1 and 24 hours, trading volume, Amihud illiquidity, Kyle’s λ (price impact of order flow), and the speed of mean-reversion (the Ornstein–Uhlenbeck half-life)—plus short lags. Edges are built from a 6-hour rolling *directed lead–lag* graph: we connect two venues if their returns are correlated, and orient the edge from the *leader* (the one that moves first) to the *follower*. The GNNs predict onset for every active, non-origin node at each snapshot, trained with class-weighted loss (positives are rare) and early stopping. Reporting a full ladder lets us attribute any gain to the right cause: the sequence model isolates the value of temporal dynamics, and the GNNs isolate the value of the network.

5 Results

The right yardstick. Depeg onset is a rare event, so we use precision–recall AUC (PR-AUC) rather than accuracy. But PR-AUC mechanically rises as the base rate rises (and our base rate grows from 2.6% at 30 minutes to 29% at 24 hours), so comparing raw PR-AUC *across* horizons is misleading. We therefore also report ROC-AUC, which is invariant to the base rate, and PR-AUC *lift* (PR-AUC minus the base rate). A model with no skill scores ROC-AUC 0.5 and lift 0.

Lead-time. Figure 2 tells the central story. At horizons of four hours or less, *no* model exceeds chance: short-term contagion onset is not predictable from price microstructure. At 24 hours, only the graph models clear ROC-AUC 0.5, with GraphSAGE reaching 0.65. Contagion onset in this market is a *day-scale* phenomenon; the hour-scale negative result is real and honestly scopes the claim.

Headline (averaged over five seeds). On the held-out SVB cluster at 24 hours, GAT attains PR-AUC 0.447 ± 0.016 , a margin of $+0.175 \pm 0.016$ over XGBoost, which sits exactly at the base rate of

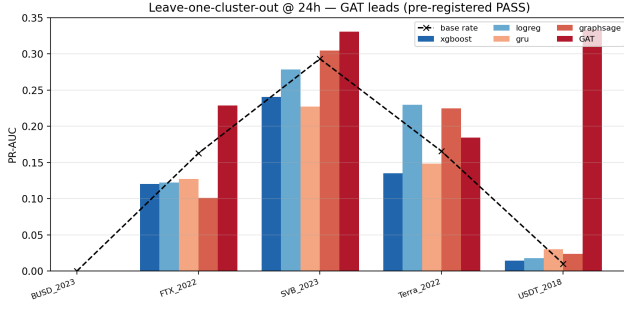


Figure 3: Leave-one-cluster-out PR-AUC at 24h. Graph models (GAT in dark red) lead the tabular baseline on the contagion-bearing folds; the single-positive USDT-2018 fold is a high-variance outlier we report but do not lean on.

Table 1: Graph ablation, held-out SVB cluster @ 24h, mean of 5 seeds. “Graph contribution” is real-edges minus no-edges, holding the architecture fixed.

Model rung	PR-AUC	graph contribution
node-only (XGBoost)	0.271	—
GraphSAGE, no edges	0.392	+0.009
GraphSAGE, real edges	0.401	
GAT, no edges	0.347	
GAT, real edges	0.447	+0.100

0.29. We stress the multi-seed reporting: a single unlucky random seed gives GAT only 0.29, so a single run would have understated the effect. In the harder LOCO protocol the per-fold margins shrink but stay positive (Fig. 3); they survive dropping the degenerate USDT-2018 fold, which contains a single positive example (GAT remains +0.083 over XGBoost on the remaining folds).

Ablation: is it the graph or the model? A skeptic will ask whether the GNN wins simply because it is a more flexible neural network, not because it uses the *network*. We answer by re-running the GNNs with all edges deleted—turning each into a per-node neural network with no message passing—and comparing to the same architecture with the real edges (Table 1). The jump from XGBoost (0.271) to the edge-free GNNs (0.35–0.39) is the value of the neural architecture. The further jump to the edge-equipped models is the value of the *graph*: the directed lead-lag edges add +0.100 PR-AUC for GAT but only +0.009 for GraphSAGE. The cross-asset relationships therefore carry genuine contagion signal, and *graph attention*—the ability to weight some neighbours more than others—is what exploits it. This both justifies the graph model and explains why attention, specifically, is the winner.

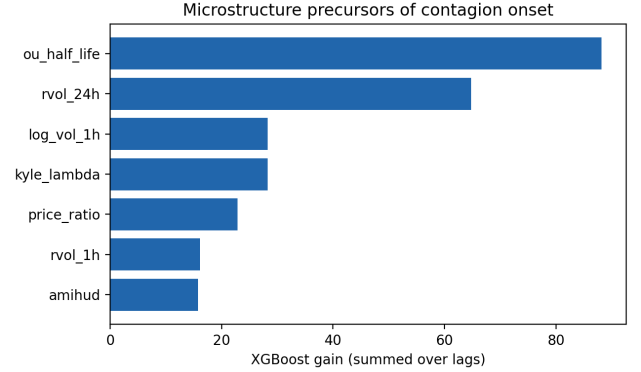


Figure 4: Which signals predict contagion onset (XGBoost gain, summed over lags). Mean-reversion speed and recent volatility dominate.

6 Interpretation and the exported hub ranking

Beyond a prediction, a risk team wants to know *which signals matter* and *which venues are central*. The strongest individual predictors (XGBoost feature importance; Fig. 4) are the speed of mean-reversion (OU half-life) and recent 24-hour volatility—intuitively, a coin that is reverting slowly and trading turbulently is fragile.

We summarize each crisis with an interpretable *hub ranking* that combines three ingredients: how much a node influences the GNN’s predictions (measured by occlusion—how much predictions change when we zero out that node, a perturbation-based and software-version-robust alternative to GNNExplainer-style masks [6]); its graph betweenness centrality; and a *propagator label* computed directly from raw prices (so it is not circular with the model). For the USDC/SVB crisis the top-ranked hub is **BUSD** (Fig. 5). We deliberately stop at *ranking*. Whether a highly ranked, highly central hub is actually *causally* responsible for spreading contagion—and what happens to a regulator who acts on a merely correlational ranking—is a distinct question that requires an intervenable model, which we address in a companion study that consumes exactly this exported ranking.

7 Limitations

We study seven episodes on the cleanly fetchable cross-asset universe of two large exchanges; two episodes are too sparse (a single active venue, almost no contagion) to carry signal. The lead-lag edges can over-credit thinly traded coins that merely appear to “lead” because of noise. Short-horizon onset is genuinely unpredictable in our data, and we report this rather than hide it. Finally, the hub ranking is correlational by construction; establishing causation is out of scope here and is the subject of companion work.

8 Conclusion

On real multi-venue data, day-scale stablecoin contagion is predictable, and a graph-attention network adds real, *edge-borne* signal—an extra 0.10 in PR-AUC over the same model without the network—while hour-scale onset is not predictable at all. The method produces an interpretable, exportable ranking of systemically central venues.

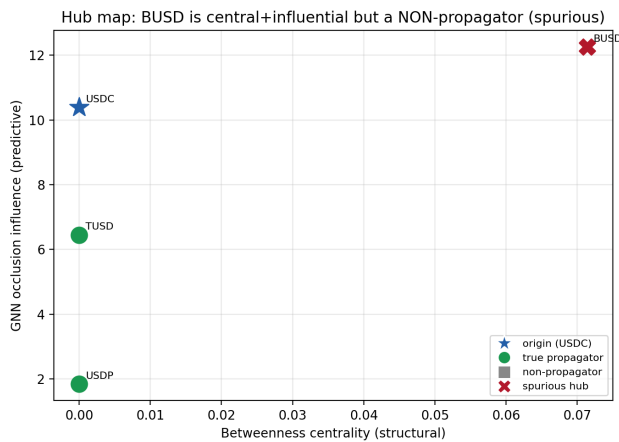


Figure 5: Exported hub map for USDC/SVB: graph betweenness (x) versus GNN influence (y), with the origin (USDC) marked separately. The top-ranked hub is the actionable output handed to causal analysis.

How to act on that ranking safely depends on whether its hubs are causal, the question we turn to next.

Reproducibility. Every number above is produced by committed scripts; a single `reproduce.sh` regenerates all datasets, results, figures, and the hub export end-to-end.

References

- [1] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proc. 22nd ACM SIGKDD*.
- [2] European Parliament and Council. 2023. Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA).
- [3] William L. Hamilton, Rex Ying, and Jure Leskovec. 2017. Inductive Representation Learning on Large Graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [4] U.S. Congress. 2025. Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act. Public Law.
- [5] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph Attention Networks. In *Proc. International Conference on Learning Representations (ICLR)*.
- [6] Rex Ying, Dylan Bourgeois, Jiaxuan You, Marinka Zitnik, and Jure Leskovec. 2019. GNNExplainer: Generating Explanations for Graph Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [7] Qingran Zhou, Eric Liu, and Alessio Brini. 2025. Graph Neural Networks for Bridge Swap Link Prediction in Uniswap v3. In *Proc. 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM. doi:10.1145/3768292.3770392